

Replication of “Medicare Accountable Care Spending Patterns: Shifting Expenditures Associated with Savings”

Eric Walczyk
PPPA 8022
12/15/2024

Introduction

Accountable Care Organizations (ACOs) within the Medicare Shared Savings Program (MSSP) aim to improve care quality while reducing costs. The original study, “Medicare Accountable Care Spending Patterns: Shifting Expenditures Associated with Savings,” sought to identify whether MSSP ACOs reallocated spending across care settings between 2013 and 2016 and how these changes contributed to financial savings. Using Data from the MSSP Public Use Files (PUFs), the authors found that reductions in inpatient and skilled nursing facility (SNF) expenditures were strongly associated with savings, while spending on physician services increased.

This replication extends the analysis using the 2014 -2017 MSSP data to test whether these findings hold over a later period. Specifically, it addresses two questions: (1) Have MSSP ACOs continued to reallocate spending across care settings? (2) Are these shifts in expenditures associated with increased savings? The replication results are compared to those of the original study to assess consistency and identify new insights.

Data and Methods

Data Source

This analysis uses MSSP PUFs for 2014-2017, which include detailed data on ACO expenditures, saving rates, and organizational characteristics. The expenditure categories analyzed include inpatient care, SNF, home health, hospice, physician services, durable medical equipment (DME), and ambulance services.

Key adjustments to the data included annualizing expenditures to account for partial-year participation and standardizing spending as percentages of total expenditures. These steps ensured comparability across ACOs of varying sizes and operational years

Analytical Models

Changes in Expenditures Over Time

To examine shifts in expenditure patterns, a fixed-effects regression was used, controlling for unobserved, time-invariant characteristics and time-specific effects:

$$\{Expenditure\ Category\}_{it} = \delta_t + \alpha_i + \mu_{it}$$

Here, $\{Expenditure\ Category\}_{it}$ represents spending as a percentage of total expenditures for ACO i in year t , δ_t captures year fixed effects, and α_i accounts for ACO-specific characteristics. This model isolates year-over-year changes.

Effects of Expenditures on Savings Rates

To determine the relationship between spending patterns and savings, another fixed-effects regression model was used:

$$\{Savings\ Rate\}_{it} = \beta_n \{Expenditure\ Categories\}_{it} + \delta_t + \alpha_i + \mu_{it}$$

In this model, $\{Savings\ Rate\}_{it}$ is the percentage of savings relative to benchmark expenditures, and β_n represents the coefficients measuring the effect of changes in specific expenditure categories on savings.

Achieving Savings

A logistic regression with fixed effects examined whether specific expenditure patterns were associated with the likelihood of achieving savings:

$$\{Saved\}_{it} = \beta_n \{Expenditure\ Categories\}_{it} + \delta_t + \alpha_i + \mu_{it}$$

Here, $\{Saved\}_{it}$ is a binary variable indicating whether ACO i achieved savings in year t . This analysis provides insights into how spending patterns differ between ACOs that saved and those that did not.

Results

The replication confirmed the trends observed in the original study, with notable shifts in spending patterns. Inpatient Care and SNF expenditures decreased as percentages of total spending, while physician services spending increased consistently, reflecting a shift towards outpatient care. The replication also showed a modest decline in home health expenditures, a trend that was not significant in the original study. These results suggest a continued reallocation of resources toward lower-cost care settings.

Changes in Expenditures Over Time

The replication confirmed most of the trends observed in the original study with notable shifts in spending patterns. As can be seen in Figure 1, SNF expenditures decreased as mean expenditures per beneficiary in an ACO, while physician spending increased consistently. The replication results differ in inpatient expenditures, where the original study found decreases every year while the replication only saw a decrease in mean spending per beneficiary in 2015.

Table 1

	Inpatient Expenditures (\$)	Skilled Nursing Facilities (\$)	Physician Services (\$)	Ambulance Services (\$)	Home Health (\$)	Durable Medical Equipment (\$)	Hospice (\$)
Year: 2015	-11.016	-11.078	65.418***	-8.145***	-8.116	9.678***	-2.517
	(26.449)	(20.836)	(12.541)	(2.321)	(10.736)	(2.286)	(6.621)
Year: 2016	64.429*	-75.837***	129.880***	-14.707***	-21.691+	-8.421***	-2.165
	(27.661)	(21.791)	(13.115)	(2.427)	(11.228)	(2.391)	(6.924)
Year: 2017	144.750***	-71.519**	231.491***	-13.462***	-15.289	-17.417***	-10.347
	(28.552)	(22.493)	(13.538)	(2.505)	(11.590)	(2.468)	(7.147)
Num.Obs.	1570	1569	1570	1570	1570	1570	1570
R2	0.939	0.915	0.981	0.919	0.938	0.868	0.901
R2 Adj.	0.899	0.860	0.968	0.866	0.898	0.783	0.838
AIC	23090.3	22326.9	20747.0	15449.7	20259.2	15402.0	18741.3
BIC	26407.4	25638.2	24064.2	18766.8	23576.4	18719.2	22058.4
Log.Lik.	-10926.150	-10545.441	-9754.523	-7105.838	-9510.617	-7082.018	-8751.638
F	23.570	16.616	78.818	17.395	23.302	10.181	14.121
RMSE	254.78	200.77	120.80	22.36	103.42	22.02	63.77

• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Effect of Expenditures on ACO Savings

The association between expenditure changes and savings was consistent with the original study. The replication found that a 1% reduction in inpatient spending was associated with a 0.54% increase in the savings rate, closely matching the original result of 0.46%. Additionally, increased spending on physician services was positively correlated with savings, though the effect size remained small. These results are further supported by the logistic regression results which showed that increases in physician expenditures and decreases in inpatient expenditures were associated with ACOs that saved.

Table 2 shows the results for both the fixed-effects and logistic models. Savings Rate 1 is the baseline fixed-effects model using savings rate as the dependent variable. Logit Model 1 is a logistic regression that uses the same independent variables as Savings Rate 1, but with a binary dependent variable indicating whether an ACO saved or did not save. Savings Rate 2 and Logit 2 exclude the ambulance and DME expenditures to address issues of multicollinearity, which is a common issue with healthcare expenditure data. One area of weakness in the replication compared to the original results is the logistic models. The original results show high statistical significance for inpatient care, hospice, SNF, and physician expenditures, as well as home health to a lesser degree of significance. My replication only shows high statistical significance for inpatient expenditures and home health to a lesser degree. I believe this is due to constraints the authors used in the original study that they did not elaborate on and that I was not able to replicate. The original study used 1377 observations for their fixed-effects model but only 624 observations for their logistic models, whereas both my models use 1569 observations for both.

Table 2

	Savings Rate 1	Savings Rate 2	Logit Model 1	Logit Model 2
Percent Inpatient	-0.550***	-0.499***	-55.861***	-54.114***
	(0.126)	(0.120)	(15.501)	(15.174)
Percent HSP	-0.350	-0.311	-30.167	-30.373
	(0.215)	(0.213)	(25.214)	(24.843)
Percent SNF	-0.357***	-0.369***	-18.271	-22.228+
	(0.102)	(0.099)	(12.193)	(11.817)
Percent PB	0.012	0.040	22.847+	23.843+
	(0.100)	(0.099)	(12.736)	(12.658)
Percent AmbPay	-0.760		-68.932	
	(0.544)		(66.119)	
Percent HHA	-0.423**	-0.380**	-37.319*	-34.401*
	(0.140)	(0.138)	(17.671)	(17.029)
Percent DME	-0.332		4064.098	
	(0.330)		(4971.058)	
Num.Obs.	1569	1569	1569	1569
R2	0.085	0.082		
R2 Adj.	-0.513	-0.515		
RMSE	2.39	2.40	0.30	0.30
• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001				

Conclusion

This replication study slightly extends the findings of the original paper, “*Medicare Accountable Care Spending Patterns: Shifting Expenditures Associated with Savings*”, by analyzing MSSP ACO data from 2014-2017. The results reaffirm some of the key conclusions of the original study while identifying some differences due to the data period and methodological constraints. Both studies confirm that MSSP ACOs reallocated spending towards lower-cost care settings, specifically inpatient and SNF expenditures declined, while physician services increased. Both studies also found strong evidence that reducing inpatient and SNF expenditures is associated with higher savings. The studies differ in their logistic model results, but both still found strong statistical significance for decreased inpatient expenditures being associated with ACOs that saved.

Policymakers and ACO administrators can use these studies to demonstrate that reducing inpatient and SNF expenditures is a successful strategy for ACOs to achieve savings. Prioritizing outpatient physician services and care coordination to reduce reliance on inpatient and post-acute care should be an area of focus for these administrators and policymakers. Future research could explore the causal mechanisms behind expenditure shifts and savings, particularly to ensure that ACOs aren't self-selecting healthier population groups to ensure they achieve savings.

This replication reinforces the broader conclusion of the original study: MSSP ACOs are demonstrating meaningful changes in spending patterns, leading to modest but impactful financial savings. The observed consistency over time underscores the potential of ACOs to drive cost-efficient care delivery while highlighting the importance of ongoing evaluation to optimize the savings strategies.

References

David B. Muhlestein P, Spencer Q. Morrison BA, Robert S. Saunders P, William K. Bleser P, Mark B. McClellan MD, Lia D. Winfield P. Medicare Accountable Care Spending Patterns: Shifting Expenditures Associated With Savings. 2018;6. Accessed November 13, 2024.
<https://www.ajmc.com/view/medicare-accountable-care-spending-patterns-shifting-expenditures-associated-with-savings>